Attempt #2

The Expression for the Denominator in the Reciprocal of the Sum of Primes

14/05/2026 - 16/05/2026

My second attempt built upon the framework that was the first attempt. This time around, instead of focusing on the reciprocal prime sum, I focused moreso on creating an expression that would add upon the first reciprocal prime sum given *n*. If there was an equation that could somehow fit in the denominator, we would have that ratio change equation we needed in Attempt #1.

$$S_p\left(n\right) = \frac{1}{2 + f\left(n\right)}$$

In order to identify an equation for *f*(*n*), I took the denominators from the S_p column (values 1-25) of the initial table and subtracted 2 from each denominator.

S_p	d
$\frac{1}{2}$	0
$\frac{1}{5}$	3
$\frac{1}{10}$	8
$\frac{1}{17}$	15
$\frac{1}{28}$	26
$\frac{1}{41}$	39
$\frac{1}{58}$	56
$\frac{1}{77}$	75

Upon generating a power regression equation for the values, lo and behold.

Power Regression for d

EQUATION

$$y = 0.647927 \cdot x^{2.29843}$$

STATISTICS

$R^2 = 1$

RESIDUALS

e_3 plot

Bingo. A regression equation with an R^2 value of 1, indicative that the regression equation is a perfect fit for all values present within the table in the *d* column.

Hence, we may formulate the following equation for $S_p(n)$.

$$S_p(n) = \frac{1}{2 + 0.647927\left(n^{2.29843}\right)}$$

Using the $P(n)$ equation generated in Attempt #1, we may notice that, although the primes are pretty close to their actual values, there must be some rounding for the generated values through $P(n)$.

n	A_p	$P(n)$
1	2	0.647927
2	3	2.5393694
3	5	4.9065673
4	7	7.5851531
5	11	10.506413
6	13	13.630078
7	17	16.929122

We can rectify this issue by only rounding the generated values from $P(n)$ with ceiling.

n	A_p	$P(n)$
1	2	1
2	3	3
3	5	5
4	7	8
5	11	11
6	13	14
7	17	17

There exists a slight odd discrepancy when *n*=4 and when *n*=6 within this sample size with range of 7, and that shows that our equation, whilst it works, is not a formula for identifying exact primes, made evident when the sample size increases.

n	A_p	$P(n)$
1	2	1
2	3	3
3	5	5
4	7	7
5	11	11
6	13	13
7	17	17
8	19	21
9	23	24
10	29	27
11	31	32
12	37	35
13	41	40
14	43	44
15	47	48

At times, the values may be off by just one integer. Increasing the sample size further increases the magnitude of deviations between the predicted values posed by P(*n*) and the actual expected values.

20	71	71
30	113	121
40	173	177
50	229	237
60	281	300
70	349	367
80	409	437
90	463	510
100	541	585

The middle column contains actual prime values at the n^{th} prime, whereas the column to the right contains generated prime values with P(*n*)

Looking at the $S_p(n)$ equation created, and comparing the values by the actual reciprocal sum, it is not difficult to see why there exists a slight fall-off between the actual prime integers and the generated prime integers.

A_{sp}	$S_p(n)$
0.5	0.5
0.2	0.2
0.1	0.1
0.0588235294	0.058823529
0.0357142857	0.035714286
0.0243902439	0.024390244
0.0172413793	0.017241379
0.012987013	0.012658228

At around $n = 8$, there is a significant deviation between A_{sp} and $S_p(n)$. In fraction form, A_{sp} is 1/77 when $n = 8$, whereas $S_p(n)$ is 1/79 when $n = 8$. This deviation translates over to our $P(n)$ equation, in which the generated prime created with $P(n)$ is offset to the actual prime number by 2.

My initial thought was that, perchance, our sample size of prime integers wasn't large enough, so I decided to increase the sample size to find a better match for $S_p(n)$.

n	P_n	S_p	d
100	541	$\frac{1}{24133}$	$24133-2$
110	601	$\frac{1}{29897}$	$29897-2$
120	659	$\frac{1}{36227}$	$36227-2$
130	733	$\frac{1}{43201}$	$43201-2$
140	809	$\frac{1}{50887}$	$50887-2$
150	863	$\frac{1}{59269}$	$59269-2$
160	941	$\frac{1}{68341}$	$68341-2$
170	1013	$\frac{1}{78149}$	$78149-2$
180	1069	$\frac{1}{88585}$	$88585-2$
190	1151	$\frac{1}{99685}$	$99685-2$
200	1223	$\frac{1}{111587}$	$111587-2$

Running the same power regression process as with the previous dataset, we get the following equation.

There is quite a substantial difference between the regression equation generated with the new larger dataset compared to the previous smaller scope dataset, however.

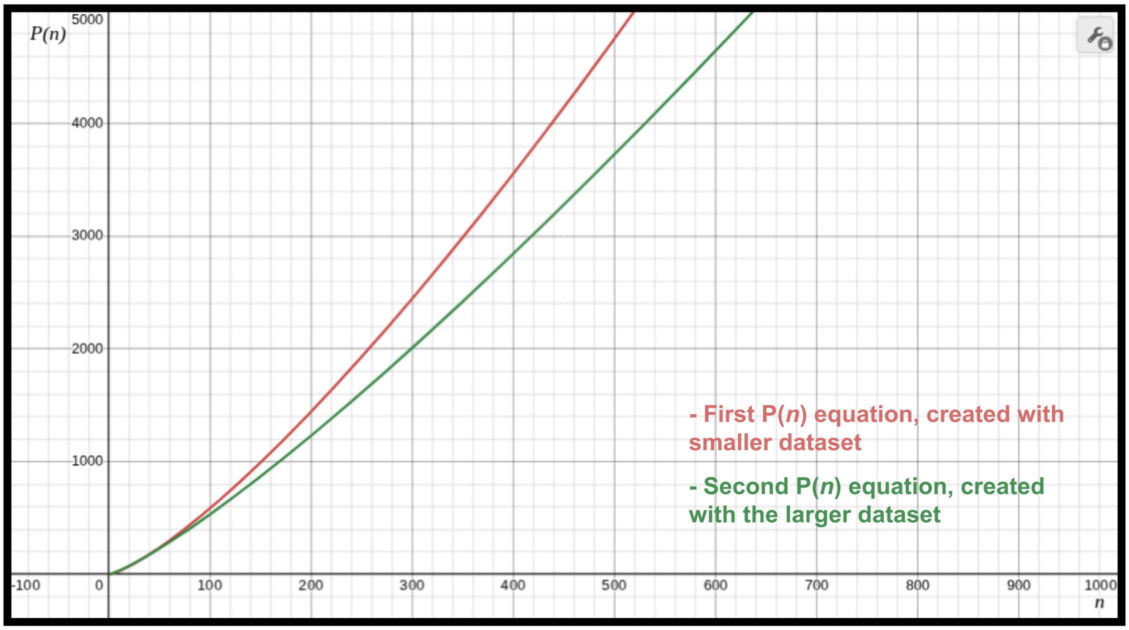
Whilst the first $S_p(n)$ equation generated was as follows:

$$S_p(n) = \frac{1}{2 + 0.647927\left(n^{2.29843}\right)}$$

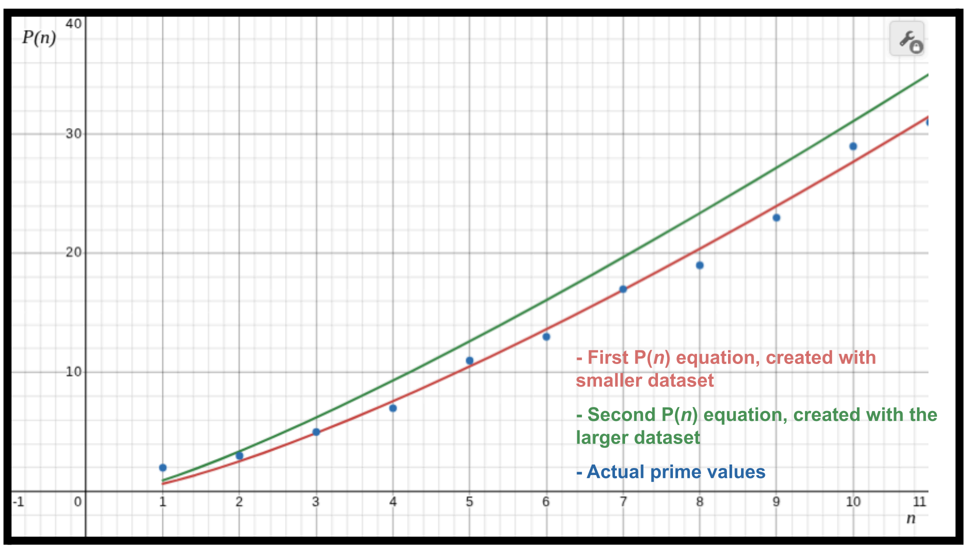
The new $S_p(n)$ equation is

$$S_p(n) = \frac{1}{2 + 0.928914\left(n^{2.20778}\right)}$$

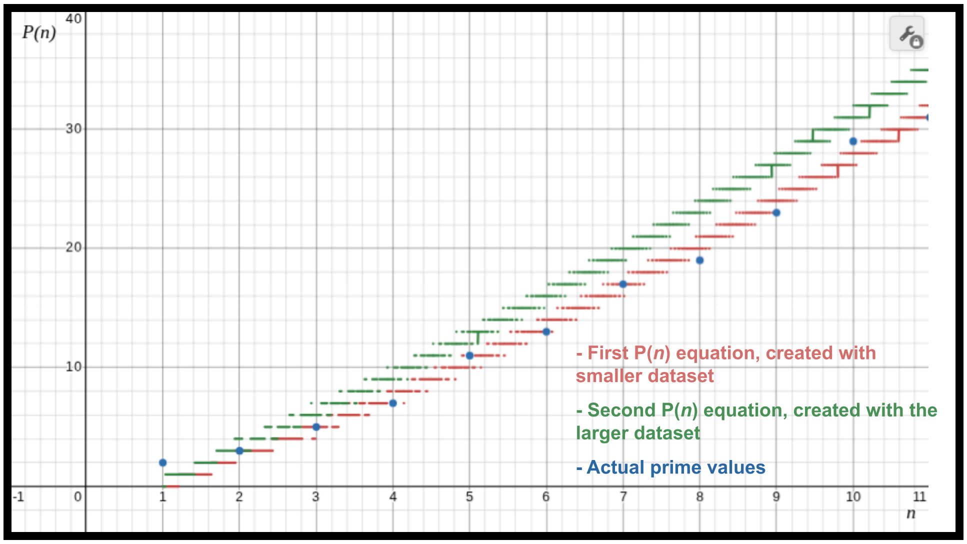
We may also begin to notice–there is a deviation between the first P(n) equation that used the first $S_p(n)$ formula, compared to the second P(n) equation that used the new $S_p(n)$ formula.



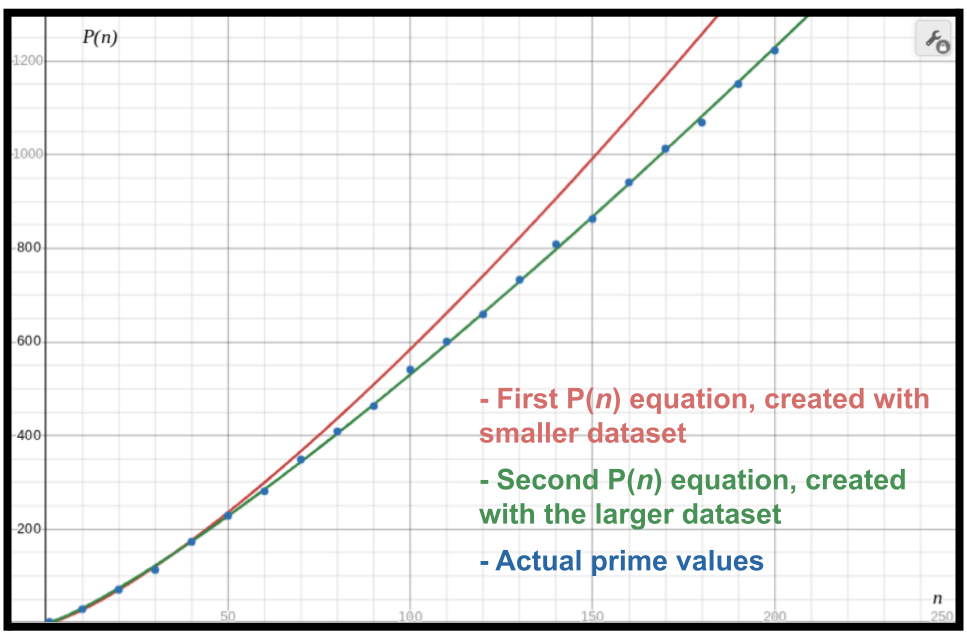
We can also compare our graphs with the actual n^{th} prime value.



But wait—previously, I also used the ceiling function with the first P(n) equation to turn continuous values into discrete integers. Using that function, we get the following graph when the ceiling function is employed both for the first P(n) equation and the second P(n) equation.



...okay, admittedly, it's confusing to interpret, but it seems as though the first P(*n*) function was closer to the initial values than the second function. However, we also need to look at the larger scope and compare the accuracies of the first P(*n*) function and the second P(*n*) function as *n* increases.



Although initially the first P(*n*) equation seemed to match the primes closer than the second P(*n*) equation, as *n* grows larger, the second P(*n*) equation seemed more accurate than the first. However, it's still not the best at generating primes.

n	A_p	$P(n)$
100	541	531
110	601	596
120	659	662
130	733	730
140	809	798
150	863	867
160	941	939
170	1013	1010
180	1069	1083
190	1151	1156
200	1223	1230

Although our generated prime values are pretty close to the actual prime integers at *n*, what I wish to attain is a formula for finding an exact prime, not a formula for finding primes close enough to their actual values, so this won't slide. A larger sample size for prime integers may be required to further advance the process of Attempt #2.